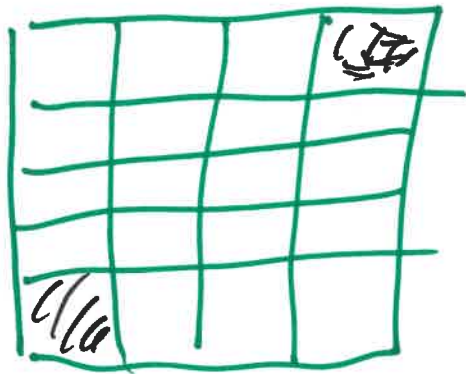


Disproof.

8×8 checkerboard
cover 2 opposite corners

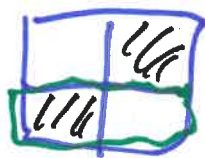


Prove there is no tiling
by dominoes

Understand Question

Try Simple Examples

2×2



Can't Fit
any domino

↑
Not allowed to use
this square

Page 12

1. The first part of the

document is a list of

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2. The second part of the

document is a list of

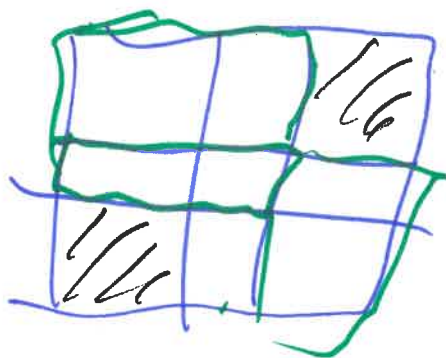
3. The third part of the



4. The fourth part of the

document is a list of

3x3



left w/ one square

$9 - 2 = 7$ squares is odd

$$7 \bmod 2 = 1$$

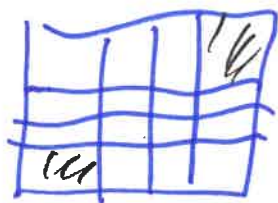
IF T covers S squares with dominos
then $|S|$ is even

$\Rightarrow$ IF $|S|$ is odd there is no domino tiling

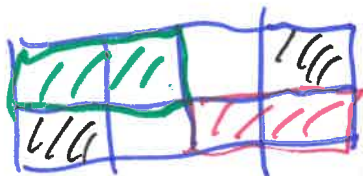
if S is even does that imply domino tiling?

No.

4x4



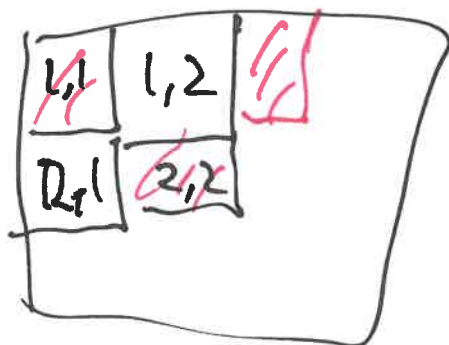
2x4



Stuck

Introduce Notation

label squares by coordinates



$$1+1 \bmod 2 = 3+1 \bmod 2$$

$$= 2+2 \bmod 2$$

even squares

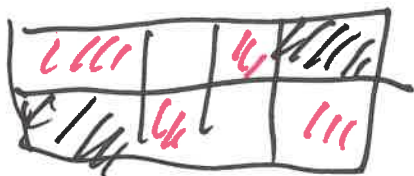
$$1+2 \bmod 2 = 2+1 \bmod 2$$

odd squares

A domino covers one odd square
one even square

$\Rightarrow$ A domino tiling T covers with
 k dominoes covers k odd squares
 k even squares
 $\# \text{ odds} = \# \text{ evens}$

No domino tiling 2×4 w/ opposite corners



4 ^{even} ~~odd~~ squares

2 odd squares

$4 \neq 2 \Rightarrow$ no tiling

$$8 \times 8 = 64 \text{ squares}$$

$$= 32 \text{ odd} + 32 \text{ even}$$

Remove 2 opposite on 8×8
both odd corners $30 \text{ odd} = 32 \text{ even}$

$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

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Lemma: IF T tiles S squares
with k dominos
then S contains k odd squares
and k even squares

Proof: Case 1: $\square$ covers x, y
and $x, y+1$

$$x+y \bmod 2 \neq x+y+1 \bmod 2$$

Covers 1 odd and 1 even

Case 2: $\square$ covers x, y
 $x+1, y$

Corollary: if S contains k_1 odd squares
and k_2 even squares
and $k_1 \neq k_2$
then S cannot be tiled by dominos

1. The first part of the paper is devoted to a general discussion of the problem.

2. The second part is devoted to a detailed analysis of the case.

3. The third part is devoted to a discussion of the results.

4. The fourth part is devoted to a conclusion.

5. The fifth part is devoted to a discussion of the results.

6. The sixth part is devoted to a conclusion.

7. The seventh part is devoted to a discussion of the results.

8. The eighth part is devoted to a conclusion.

9. The ninth part is devoted to a discussion of the results.

10. The tenth part is devoted to a conclusion.

11. The eleventh part is devoted to a discussion of the results.

12. The twelfth part is devoted to a conclusion.

13. The thirteenth part is devoted to a discussion of the results.

A set S has property X
 $S \subset \mathbb{Z}$

if for any $n \in \mathbb{N} \exists x \in X$
such that $x > n$

S has no bounds
 S is unbounded



Suppose $S = A \cup B$

and S has property X

then either A or B (inclusive)
or
has property X

— Add hypothesis if confused
try to remove later

$A \cap B = \emptyset$ A and B are disjoint
Worst Case. Remove $(A \cap B)$
 $B -$

The first part of the book is devoted to the study of the properties of the function $f(x)$ which is defined on the interval $[0, 1]$ and satisfies the conditions

$f(0) = 0$, $f(1) = 1$, $f(x) \geq 0$, $f(x) \leq 1$, $f(x) + f(1-x) = 1$.

In the second part of the book the author considers the problem of the construction of the function $f(x)$ which satisfies the conditions

$f(0) = 0$, $f(1) = 1$, $f(x) \geq 0$, $f(x) \leq 1$, $f(x) + f(1-x) = 1$, $f(x) = f(1-x)$.

In the third part of the book the author considers the problem of the construction of the function $f(x)$ which satisfies the conditions

$f(0) = 0$, $f(1) = 1$, $f(x) \geq 0$, $f(x) \leq 1$, $f(x) + f(1-x) = 1$, $f(x) = f(1-x)$, $f(x) = f(x^2)$.

In the fourth part of the book the author considers the problem of the construction of the function $f(x)$ which satisfies the conditions

$f(0) = 0$, $f(1) = 1$, $f(x) \geq 0$, $f(x) \leq 1$, $f(x) + f(1-x) = 1$, $f(x) = f(1-x)$, $f(x) = f(x^2)$, $f(x) = f(x^3)$.

Proof by Contradiction,

Assume False.

$$S = A \cup B$$

S has property X

A does not have X

B does not have X

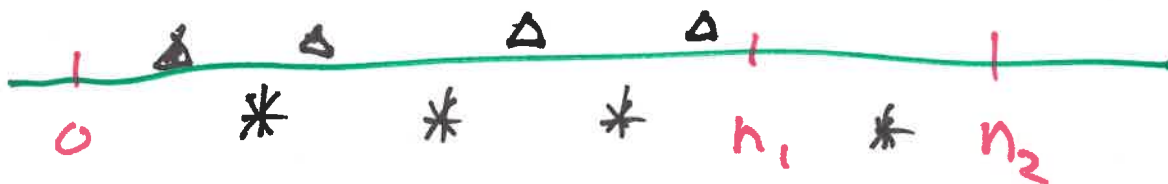
Rely on
excluded
Middle
 $\text{Not}[F] = T$

$$\neg(p \vee q) = \neg p \wedge \neg q$$

$$\exists n_1 \text{ s.t. } \forall x \in X \quad x \leq n_1$$

$$\exists n_2 \text{ s.t. } \forall x \in X \quad x \leq n_2$$

$$\begin{pmatrix} \Delta \in A \\ * \in B \end{pmatrix}$$



$n = \max(n_1, n_2)$ is a bound for S

if $x \in S$ then either $x \in A$ or $x \in B$

Case 1: $x \in A \Rightarrow x \leq n_1 \leq n$

Case 2: $x \in B \Rightarrow x \leq n_2 \leq n$

Conclude $x \leq n$
 S does not have X

polysyllabic and free

9257 222

X was for 2005 A

W. Michaelson and S.

$$\int \sin q' = -\cos q' + C$$

The map shows the northern Adriatic coastline of Italy. Sampling stations are indicated by numbers 1 through 10. Station 1 is located near the Gulf of Genoa. Stations 2 through 10 are distributed along the coast from Liguria to the Marche region. The map includes latitude lines (44°N, 45°N) and longitude lines (10°E, 12°E, 14°E). A scale bar at the bottom indicates distances from 0 to 100 km.

Figure 1 illustrates the stages of nest building and egg incubation. The sequence shows a bird sitting on a nest made of sticks, with the number of eggs increasing from one to ten, and finally showing a fledgling bird.

Lemma! $Y \Rightarrow Z$

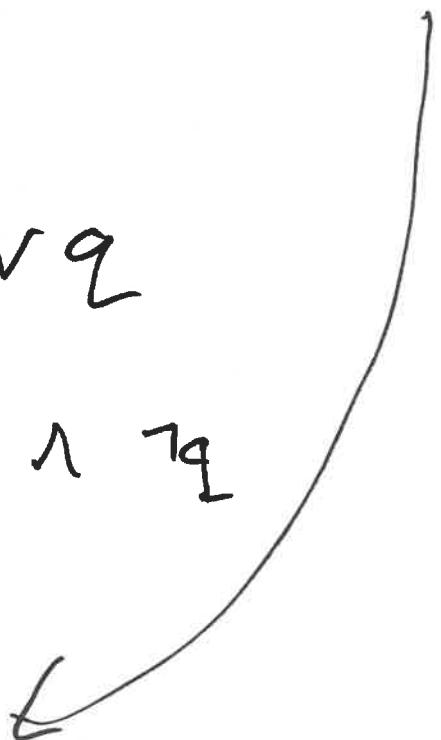
Proof by Contradiction

$$\neg (Y \Rightarrow Z) = Y \wedge \neg Z$$

Assume 

$$Z = p \vee q$$

$$\neg Z = \neg p \wedge \neg q$$


$$Y \wedge \neg p \wedge \neg q$$

Move to work with $\neg$ for a proof

